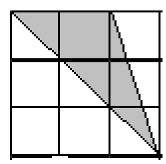
JMO – 2006

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

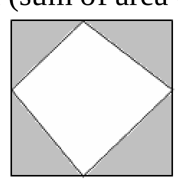
1. Out of 50 student of a class 38 playing cricket, 29 playing football and 23 playing both. How many students not playing both the games?
2. If sum of first 100 positive integers is 5050, then find the sum of first 100 positive odd integers.
3. In the figure 3X3 square divided into 9, 1X1 squares. Find the area of the shaded portion.



4. I have traveled 80 km by bicycles. First 4 hours I traveled with half of my speed and rest 4 hours in my usual speed. Then find my speed per hour.
5. If we represent $\frac{1}{7}$ in decimal number, then after decimal what is 1985^{th} digit?
6. Sides of a 3X3X3 cube painted with a colour. Then it divided into 27 cubes of 1X1X1. From each small cubes how many cubes are painted with two sides?
7. Number of seconds in 2 hours is same as number of minutes in how many days?
8. A, B and C are non zero numbers. Find A, B and C where

$$\begin{array}{r} BB \\ +\ BB \\ \hline ABC \end{array}$$

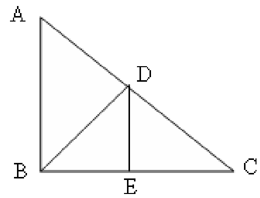
9. Find the quotient if LCM of first 40 consecutive positive integers is divided by the LCM of first 30 consecutive positive integers.
10. Which is nearest to 0.62? $\frac{6}{10}, \frac{7}{11}, \frac{10}{17}, \frac{11}{18}$.
11. In the figure four isosceles right triangles arranged to form a square. If area of white portion is 18 sq units, then find the area of shaded portion (sum of area of four isosceles right triangles).



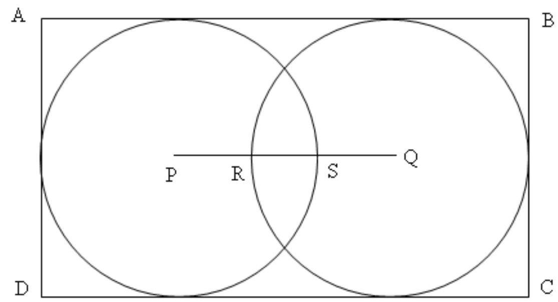
12. If in any year there are 5 Sundays in the month of February, then the same year name the day of February 14.
13. Which fractions are in ascending order?

    (i) $\frac{7}{11}, \frac{5}{8}, \frac{3}{5}, \frac{2}{3}$ (ii) $\frac{2}{3}, \frac{3}{5}, \frac{5}{8}, \frac{7}{11}$ (iii) $\frac{3}{5}, \frac{5}{8}, \frac{2}{3}, \frac{7}{11}$ (iv) $\frac{3}{5}, \frac{5}{8}, \frac{7}{11}, \frac{2}{3}$
14. In a four wheeler each type of tyre can be used upto 10000 km. If in that vehicle we have to travel 15000 km then how many extra tyre we should take to complete the journey?

15. In a market two types of mangoes was selling. One is at Rs 31 per kg and other at Rs 29 per kg. What quantity of mangoes from each type one can purchase from Rs. 459.
16. In the figure $m\angle ABC = m\angle BDC = m\angle DEC = 90^0$. If $m\angle BCA = 27^0$, then find $m\angle BDE = ?$



17. Which number should be added to 7 and 22 to make each one as a perfect square?
18. ABCD is a rectangle. P and Q are centers of the circle. Radius of each circle is 6 cm, RS = 2 cm. Find area of rectangle ABCD.



19. How many different numbers can be written as sum of two numbers of the set {1, 3, 5, 7, 9, 11, 13}.
20. Out of {-16, -4, 0, 2, 4, 12} which two numbers having largest subtraction result?
21. Sum of two positive integers is 11 and product is 24. Find the numbers.
22. For the increase of 3^0 in temperature volume of the gas increased by 4 cubic cm. If at temperature 32^0, volume of the gas was 24 cubic cm, then at 20^0 temperature what was the volume of the gas?
23. In an Mathematics examination there are 75 questions out of which 10 questions are verbal maths, 30 questions are of algebra and 35 questions are of geometry. Gajanan given 70% of verbal maths, 40% of algebra and 60% of geometry correct answer, still he gave less than 60% of correct answer in mathematics. How many more questions he can give right answer so as to give 60% correct answer in mathematics.
24. In a straight road at one side 6 trees are there with equal distances. If the distance between first and fourth is 60 meters, then find distance between first and last.
25. Bulu, Chintak, Fatima, Jatin and Mamata each have some amount of money. Jatin and Bulu both having less amount of money than Fatima. Bulu and Chintak having more money than Mamata. Jatin having more money than Mamata but less money than Bulu. Then who among all have less money?

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